Pintoo Kumar

Backend Engineer | Golang, GraphQL, Docker, Kubernetes

Professional Summary

Backend Engineer with 7+ years of experience, including 3+ years specializing in Golang and distributed backend systems. Proven track record in building scalable, containerized services using **Go (Gin, gRPC, GORM)**, **GraphQL (Apollo, gqlgen, subscriptions)**, and **PostgreSQL**. Skilled in designing and deploying microservices on **Docker** and **Kubernetes**, implementing realtime features, and optimizing APIs for high-volume traffic. Hands-on with **Supabase (Auth, Edge Functions, Realtime)** and AI-powered integrations using **OpenAI GPT-4o, Claude 3, Cursor, v0.dev, Lovable**. Strong problem-solving ability with deep knowledge of system design and clean code practices.

Core Skills

- **Languages & Frameworks:** Golang (Gin, gRPC, GORM), TypeScript, Python 3, Node.js, NestJS, FastAPI
- **APIs & Protocols:** REST, GraphQL (Apollo Server/Client, gqlgen, Subscriptions), WebSockets
- **Databases:** PostgreSQL, Supabase, MongoDB, Redis
- **Cloud & DevOps:** Docker, Kubernetes, Helm, Azure AKS, AWS, DigitalOcean, GitHub Actions, Jenkins
- **AI Tools & Integrations:** OpenAI GPT-4o, Claude 3, v0.dev, Cursor, Lovable, LLM-powered automation
- **Testing & Quality:** Jest, Cypress, Mocha, Chai, Prometheus, Grafana
- **Other:** Supabase Auth, Edge Functions, Realtime, CI/CD, Agile

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Selected Projects

Supply Chain Management System

A blockchain-based vaccine shipment tracker system that monitored real-time temperature and geolocation during transit.

- Developed blockchain infrastructure using **Hyperledger Fabric**, **Shell scripts**, and **Docker** for network orchestration.
- Implemented smart contracts (chaincode) in **Golang** and stored on-chain data in **CouchDB**.
- Built **Node.js** REST APIs to facilitate blockchain interaction and **MongoDB** for off-chain storage.
- Created real-time UI with **React.js**, **Redux**, **TailwindCSS**, and **TypeScript**.
- Integrated backend/frontend using **Axios**, and implemented unit testing via **Jest**.

DeFi Lending Platform

A blockchain-powered decentralized finance platform enabling tokenized lending and collateralized borrowing.

- Architected secure blockchain interactions via **Hyperledger Fabric** and implemented business logic in **Golang** smart contracts.
- Set up decentralized network orchestration using **Shell scripting**, **Docker containers**,

and **CouchDB** for on-chain persistence.
- Developed backend services with **Node.js**, RESTful APIs, and leveraged **MongoDB** for fast off-chain record management.
- Built responsive, single-page UI components using **React.js**, **Redux**, **TailwindCSS**, and **TypeScript**, connecting via **Axios** to blockchain APIs.
- Ensured consistent application quality through automated testing in **Jest** and Git-managed CI workflows.
- Added AI-powered automation using OpenAI GPT-4o and Claude 3.
- Implemented Supabase Edge Functions for event-driven workflows.

E-Commerce Platform

- Architected backend with **Golang (Gin, GORM)** and GraphQL APIs for inventory, payments, and CRM modules.
- Integrated Stripe and Razorpay; optimized PostgreSQL queries for high concurrency.
- Deployed with Docker/Kubernetes and implemented CI/CD pipelines.

Web3 Backend APIs
Backend services for a decentralized NFT marketplace and token issuance system.

- Developed scalable REST and GraphQL APIs using **NestJS** and **TypeScript**.
- Created blockchain interactions using **ethers.js** to connect to EVM-based contracts.
- Managed transaction creation, verification, and wallet-based role logic.
- Wrote **unit**, **integration**, and **e2e tests** using **Jest**.
- Documented all routes and payloads using **Swagger**.
- Handled schema migrations and connection pooling with **TypeORM** and **PostgreSQL**.

Title-Direct
A real estate and mortgage platform for title management and advisory services.

- Built secure backend services using **NestJS**, **GraphQL**, **TypeScript**, **Prisma**, and **PostgreSQL**.
- Integrated blockchain ledger using **R3 Corda** and implemented scalable infrastructure with **Docker**, **Kubernetes**, **Helm**, and **Azure Kubernetes Service (AKS)**.
- Managed authentication with **Azure Active Directory**, created custom workflows, and implemented monitoring using **New Relic**.
- Utilized **Swagger** (OpenAPI) for API documentation and **Boomi** for enterprise integrations.
- Designed UI components in **Next.js** with **Material UI (MUI)** and managed real-time state with **Redis**.
- Conducted rigorous testing using **Jest** and **Cypress** for unit and end-to-end validation.

DocVenture
A digital document automation platform for secure contract generation and approvals.

- Developed full-stack components using **NestJS**, **GraphQL**, **PostgreSQL**, and **Prisma**.
- Built AI-powered automation features with **OpenAI**, **LLMs**, and integrated backend logic in **TypeScript**.
- Designed frontend in **Next.js** and **MUI** with real-time interaction via **Apollo Client**.
- Utilized various **Azure services** for deployment, storage, and identity management.
- Managed API documentation with **Swagger**, and enabled continuous deployment via **GitHub Actions**.
- Ensured reliability through observability tools like **Prometheus** and **Grafana**.

Education & Certifications

- B.Tech in Electrical Engineering – Krishna Institute of Engineering & Technology, 2015
- **Docker & Kubernetes: The Complete Guide** – Udemy
- **Full Stack Web Development with Node.js and React** – Udemy